

Satvik Singh

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Backend Software Engineer (3 yrs) specializing in Go microservices and distributed systems. Founded two backend services from scratch and owns a risk-scoring engine processing ~500M events/month across 6 production regions — from data model and event-driven APIs to reliability and on-call.

SKILLS

- **Languages:** Go, Java, TypeScript/JavaScript, Python, SQL
- **Backend:** Microservices, distributed systems, REST APIs, event-driven (Kafka/MSK, RabbitMQ), Temporal workflows, Spring Boot, multi-tenancy, ABAC authorization, Unleash feature flags
- **Data:** PostgreSQL, MySQL, Cassandra, DynamoDB, MongoDB, Redis
- **AI/ML:** Retrieval-augmented generation (RAG), vector/semantic search (AWS OpenSearch Serverless), embeddings, LLM integration (OpenAI GPT-4o/4.1, AWS Bedrock/Claude), prompt engineering, LLM guardrails
- **Infra & Observability:** Docker, AWS (ECS/MSK/STS), CI/CD, Observe, Datadog, Git

EXPERIENCE

Safe Security — Software Development Engineer

Remote · Jan 2025 – Present

SAFE One — enterprise cyber-risk quantification (CRQ) & third-party risk management (TPRM) platform; Go microservices, multi-tenant, 6 production regions.

- **Build and operate the platform's risk-scoring engine** — the system that computes every customer's cyber-risk posture — **processing ~500M events/month across 6 production regions** and serving **~1.16M risk evaluations/month at p99 585 ms**. Hardened it through a 3-week reliability push that resolved **10 critical-memory + 14 CPU incidents** via a memory-retention fix, database indexing, cache-TTL redesign, a load-shedding kill-switch, and a self-healing job that auto-regenerates stale risk data every 10 minutes — and drove a platform-wide push that brought **P95 latency of critical (P0) customer APIs under 1 second**.
- **Designed and shipped a customer-facing executive risk-reporting product** that unifies five risk dimensions — configuration posture, contract gaps, assessment severity, risk scores, and live activity — into one drill-down dashboard, giving CISOs a single view of their third-party risk; the aggregation backend handles **~109M events/month** and **~392K requests every two weeks**.
- **Built a secure cross-tenant data-exchange service from scratch to production** (cross-account AWS access via STS, cost-optimized single-table NoSQL) that moves sensitive risk data between customer environments with **zero error-budget breaches in 90 days** — owning architecture, event-driven processing, multi-region deployment, and CI/CD.
- **Built and own the evidence/document-analysis service behind the platform's AI capabilities:** secure upload and malware scanning, then an event-driven pipeline that chunks and embeds uploaded documents into a **retrieval-augmented generation (RAG)** index (**AWS OpenSearch Serverless**, hybrid vector + keyword search) — powering **LLM summarization/classification of evidence and auto-answering of security questionnaires from documents** (OpenAI GPT-4o/4.1 and Claude via AWS Bedrock through an internal multi-provider gateway).
- **Own a workflow-automation platform running ~147K automated workflows/month;** building a generic HTTP-integration node that lets teams wire any internal or external API into a workflow without custom code — turning a fixed workflow tool into an extensible integration platform.
- **Built enterprise-grade access control (attribute-based) with organization-hierarchy data scoping** across the platform, letting large customers segment risk data by business unit and role — a P0 requirement that unblocked deals for global-manufacturing and Fortune-500 healthcare enterprises needing strict data isolation.
- **Automated the third-party assessment lifecycle end-to-end** — auto-assessment and reminder/escalation — on an event pipeline peaking at **500K+ messages**, removing manual follow-up for risk analysts and speeding assessment turnaround.

Paisabazaar — Software Development Engineer

Gurugram, India · Jul 2023 – Jan 2025

India's largest digital consumer-credit marketplace.

- **Cut response time ~80%** on a core visit-ID service via a failsafe mechanism (~10% undelivered OTPs). *Node.js, Python, Redis, Kafka, MySQL, Cassandra.*
- Built a **company-wide authN/authZ Java package** for microservices with request monitoring. *Java, Servlet, Redis, LRU caching.*
- Worked on the **RBI Account Aggregator** stack — onboarded a data provider and built user-consent workflows + core modules (gateway, webhook handling, consent state). *Spring Boot, Node.js, MySQL, Cassandra, PostgreSQL.*
- Built a **rate limiter** (~10% dropped requests) and internal Voucher/Refund tooling (~20% processing time). *Node.js, Redis, React, MySQL.*

PROJECTS

- **InvestIQ — AI Investment Platform** (*personal*): Flask + React app with an **automated trading engine** (Upstox) — 6-layer risk validation, GTT/trailing stops, circuit breaker, kill switch; an **XGBoost + RandomForest ensemble** with walk-forward validation + Monte Carlo; **FinBERT** sentiment; **HRP** portfolio optimization; multi-LLM chat (Gemini/Ollama). Dockerized PWA. *Python, Flask, React, scikit-learn, XGBoost, Docker.*

EDUCATION

National Institute of Technology (NIT), Meghalaya — B.Tech, Electrical & Electronics Engineering

Jun 2019 – Jun 2023

CGPA 9.0